

Biological actions of curcumin on articular chondrocytes.



OBJECTIVES: Curcumin (diferuloylmethane) is the principal biochemical component of the spice turmeric and has been shown to possess potent anti-catabolic, anti-inflammatory and antioxidant, properties. This article aims to provide a summary of the actions of curcumin on articular chondrocytes from the available literature with the use of a text-mining tool. We highlight both the potential benefits and drawbacks of using this chemopreventive agent for treating osteoarthritis (OA). We also explore the recent literature on the molecular mechanisms of curcumin mediated alterations in gene expression mediated via activator protein 1 (AP-1)/nuclear factor-kappa B (NF-kappaB) signalling in chondrocytes, osteoblasts and synovial fibroblasts. **METHODS:** A computer-aided search of the PubMed/Medline database aided by a text-mining tool to interrogate the ResNet Mammalian database 6.0. **RESULTS:** Recent work has shown that curcumin protects human chondrocytes from the catabolic actions of interleukin-1 beta (IL-1beta) including matrix metalloproteinase (MMP)-3 up-regulation, inhibition of collagen type II and down-regulation of beta1-integrin expression. Curcumin blocks IL-1beta-induced proteoglycan degradation, AP-1/NF-kappaB signalling, chondrocyte apoptosis and activation of caspase-3. **CONCLUSIONS:** The available data from published in vitro and in vivo studies suggest that curcumin may be a beneficial complementary treatment for OA in humans and companion animals. Nevertheless, before initiating extensive clinical trials, more basic research is required to improve its solubility, absorption and bioavailability and gain additional information about its safety and efficacy in different species. Once these obstacles have been overcome, curcumin and structurally related biochemicals may become safer and more suitable nutraceutical alternatives to the non-steroidal anti-inflammatory drugs that are currently used for the treatment of OA. Copyright 2009 Osteoarthritis Research Society International. All rights reserved.